GreenhouseTwin n79

The greenhouse–cucumber model without flags:
climate, photosynthesis, crop growth and LAI

Greenhouses project — model reference

Branch `n79`, 21 October 2026

Abstract

This document specifies the model coded in `invernaderosN79/GreenhouseTwin`: one equation set, no behaviour flags. It is the configuration chosen as the baseline for making the physics more descriptive: the V2.2 engine with the n79 settings (fixed compartment geometry, energy-conserving solar column, cover node with condensation, open-loop logged vents, stack leakage, closed floor balance, internal mass, full lamp heat, crop respiration) and Aaron Vélez's generic cucumber model with its canopy and LAI extension (the n105 crop set-up). Every equation is written with the term names of the code (`src/climate.jl`, `src/crop.jl`), so each line can be checked against `model_terms`. Parameter tables are generated from the parameter files. Known structural gaps are listed at the end; they are the starting point of the physics work.

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1 Overview

Scope. One compartment of the Autonomous Greenhouse Challenge 2018 (AiCU, Bleiswijk; 96 m^2, Venlo, gutter 5 m), cucumber crop transplanted on 14 August 2018, 115 simulated days. All quantities are per m^2 of floor.

Two models, offline coupling.

- The **climate model** is a system of 8 ODEs integrated day by day (Tsit5, 5-min output):

$$u = \left[T_1,\ T_2,\ C_1,\ V_1,\ e_{\text{int}},\ T_5,\ T_d,\ T_m \right], \tag{1}$$

  canopy temperature, air temperature, CO_2 concentration [mg m^{-3}], vapour pressure [Pa], integral of the heating error, floor layer, deep soil and internal mass temperatures [K].
- The **crop model** is a daily step (Section 3) driven by measured climate. Its daily LAI, non-leaf respiration and standing fresh mass are inputs of the climate model (`test/crop_series.jl`). The crop does not see the simulated climate (live coupling is a planned extension).

Provenance and verification. The code is GreenhouseSim V2.2 (git tag `v2.2-flags-final`) with every flag resolved to its n79 value and dead branches removed. `test/equivalence_synthetic.jl` and `test/equivalence_n79.jl` compare it with V2.2 run with the n79 flags (Section 7).

Inputs. Table 1. Weather and LAI are functions of absolute time; controls and crop inputs of time since the season start. All are 5-min step functions.

2 Climate model

2.1 State equations

With $c_{\text{air}} = \phi_2 \rho_3 \alpha_5$ the air heat capacity per m^2 floor and $m_{\text{air}} =$ `air_cap_mult`,

$$\alpha_1 L\, \dot{T}_1 = r_1 + Q_{lc} + r_6 - h_1 - l_1 - r_7 - r_{14}, \tag{2}$$

$$m_{\text{air}}\, c_{\text{air}}\, \dot{T}_2 = h_1 + h_2 + h_4 - h_7 - h_{11} - r_{12} - r_{13} - l_2 + Q_{sa} - Q_{af} - Q_{am} + Q_{la}, \tag{3}$$

$$\phi_2\, \dot{C}_1 = o_1 + o_2 + o_3 - A - o_5 + o_R, \tag{4}$$

$$\kappa_3\, \dot{V}_1 = p_1 + p_2 + p_3 + p_4 - p_5 - p_6 - p_7 - p_8 - p_9 - p_{10}, \qquad \kappa_3 = \frac{\psi_1 \phi_2}{\omega_2 T_2}, \tag{5}$$

$$\dot{e}_{\text{int}} = \text{(anti-windup integral of } U_{11} - T_2\text{, eq. 41)}, \tag{6}$$

$$C_{\text{soil}}\, \dot{T}_5 = Q_{af} - Q_{fd} + h_{11} + r_{14} - r_{11} + Q_{fs}, \tag{7}$$

$$\dot{T}_d = (T_5 - T_d)/\tau_d, \tag{8}$$

$$\left(C_m + c_{p,\text{crop}}\, \text{FW}\right) \dot{T}_m = Q_{am} + Q_{ms}. \tag{9}$$

The cover temperature T_c and the screen temperature $T_{\text{scr}} = \frac{1}{2}(T_o + T_2)$ are algebraic. Every term is defined below with the name it has in `model_terms`.

Table 1: Model inputs (`src/inputs.jl`; AGC-2018 construction in `src/agc_data.jl`).

symbol	code	meaning (AiCU source)
I_{glob}	`Iglob`	global radiation outside [$\mathrm{W\,m^{-2}}$] (meteo)
T_{sky}	`Tsky`	sky temperature from the pyrgeometer [K]
T_o, v, V_o	`Tout, Wind, VPout`	outside temperature, wind speed, vapour pressure
C_o, T_{mc}	`CO2out, TmechCool`	outside CO_2 834.7 $\mathrm{mg\,m^{-3}}$; mechanical-cooling surface 273.15 K
L	`LAI`	leaf area index per $\mathrm{m^2}$ floor (crop series)
U_1, U_{bl}	`EnScr, BlackScr`	energy and blackout screen closure
U_{lee}, U_{win}	`VentLee, VentWind`	leeward, windward roof-vent openings
U_{10}	`CO2Valve`	CO_2 valve opening (setpoint rule scaled to the logged daily dosage)
U_{11}	`HeatSP`	heating setpoint [K]
U_{12}	`Lamps`	lamp fraction
T_{log}	`Tpipe`	logged pipe temperature [K] (both circuits folded; no reading $\to T_o + 2$)
R_{crop}, FW	`CropResp, CropFW`	crop respiration at 20 °C [$\mathrm{mg\,CO_2\,m^{-2}\,s^{-1}}$]; standing fresh mass [$\mathrm{kg\,m^{-2}}$]
$U_{2\ldots7}$, U_9	`FanPad`, `MechCool`, `Heater`, `Shade`, `SideVent`, `ForcedVent`, `Fog`	other actuators; zero in the AiCU season

2.2 Solar radiation: the energy-conserving column

For each band b (PAR $I_P = \eta_2 I_{\text{glob}}$, NIR $I_N = \eta_3 I_{\text{glob}}$, $\eta_2 + \eta_3 = 1$) five elements are stacked from the sky down: roof, energy screen, blackout screen, canopy, opaque floor. Each layer is symmetric with transmission t and reflection r:

$$\text{roof:}\quad t_r = (1-\eta_1)\tau_1\ (=0.65),\qquad r_r = 1 - t_r - \eta_1 - a_c\ (=0.15), \tag{10}$$

$$\text{screen at closure } U:\quad t = 1 - U(1-\tau_s),\qquad r = U\rho_s\qquad \text{(Vanthoor 2011, 8.14–8.17)}, \tag{11}$$

$$\text{canopy:}\quad t_c = e^{-kL},\qquad r_c = \rho_\infty(1-t_c)\qquad \text{(Vanthoor 8.31–8.32)}, \tag{12}$$

with $k = \beta_1$, $\rho_\infty = \rho_1$ (PAR) and $k = \beta_3$, $\rho_\infty =$ `rho_can_nir` (NIR); floor reflectance ρ_2 (PAR), $1-\alpha_7$ (NIR); screens: energy $(\tau_s, \rho_s) = (0.75, 0.15)$, blackout $(0.02, 0.50)$, NIR = PAR (provisional). With all repeated reflections (adding method), from the bottom up,

$$R_4 = r_4 + \frac{t_4^2\rho_f}{1 - r_4\rho_f},\quad R_j = r_j + \frac{t_j^2 R_{j+1}}{1 - r_j R_{j+1}}\ (j = 3,2,1), \tag{13}$$

$$D_2 = \frac{t_1}{1 - r_1 R_2},\quad D_{j+1} = \frac{t_j D_j}{1 - r_j R_{j+1}}\ (j = 2,3),\quad D_5 = \frac{t_4 D_4}{1 - r_4\rho_f}, \tag{14}$$

$$A_1 = (1 - t_1 - r_1)(1 + R_2 D_2),\quad A_j = (1 - t_j - r_j)(D_j + R_{j+1}D_{j+1}),\quad A_4 = (1 - t_4 - r_4)(D_4 + \rho_f D_5), \tag{15}$$

$$A_f = (1 - \rho_f)D_5,\qquad R_1 + A_1 + A_2 + A_3 + A_4 + A_f = 1. \tag{16}$$

Booking [$\mathrm{W\,m^{-2}}$]: canopy $r_1 = Q_{\text{can}} = I_P A_4^P + I_N A_4^N$; floor level $Q_{\text{flr}} = I_P A_f^P + I_N A_f^N$, of which the floor gets $Q_{fs} = (1 - f_m)Q_{\text{flr}}$ and the internal mass $Q_{ms} = f_m Q_{\text{flr}}$; roof absorption $Q_{\text{roof}} = \sum_b I_b A_1^b$ splits into structure $Q_{st} = \sum_b I_b A_1^b \min(\eta_1/(1 - t_r - r_r), 1)$ (to the air) and glass (to the cover); screen absorption $Q_{\text{scr}} = \sum_b I_b (A_2^b + A_3^b)$ goes `scr_abs_air` to the air and the rest to the cover:

$$Q_{sa} = Q_{st} + s_a Q_{\text{scr}},\qquad Q_{\text{cov}} = (Q_{\text{roof}} - Q_{st}) + (1 - s_a)Q_{\text{scr}},\qquad Q_{\text{can}} + Q_{\text{flr}} + Q_{sa} + Q_{\text{cov}} + Q_{\text{out}} = I_{\text{glob}}. \tag{17}$$

2.3 Plant light and photosynthesis

Stomata and photosynthesis see the PAR on both canopy faces plus lamp PAR:

$$\mathrm{PAR}_p = I_P D_4^P + I_P \rho_2 D_5^P + P_{\mathrm{lamp}}\,\mathrm{clamp}(U_{12},0,1) \qquad [\mathrm{W\,m^{-2}}], \tag{18}$$

(P_{lamp} = `lamp_par_W`). Canopy assimilation (FvCB, constant effective conductance $g = g_s g_m/(g_s + g_m)$) is

$$A = 44010\,\frac{1-e^{-k_p L}}{k_p}\,\min(A_j, A_c, A_p) \quad [\mathrm{mg\,CO_2\,m^{-2}s^{-1}}], \qquad k_p = 0.7, \tag{19}$$

evaluated at T_2, C_1, PAR_p with:

$$I = 4.6\times10^{-6}\,\mathrm{PAR}_p, \quad C_i = 0.509\times10^{-6} C_1, \quad \Gamma^* = \frac{0.5\,O_a}{S_{c/o,25}\exp\!\big(\frac{T-298.15}{T}\frac{E_{Soc}}{298.15R}\big)}, \tag{20}$$

$$J = \frac{(\alpha I + J_{\max}) - \sqrt{(\alpha I + J_{\max})^2 - 4\theta\alpha I J_{\max}}}{2\theta}, \quad V_{c\max} = \frac{V_{c\max,25}\,Q_{10,V}^{0.1(T-298.15)}}{1+e^{0.128(T-315.15)}}, \tag{21}$$

$$K_C = K_{C25} Q_{10,KC}^{0.1(T-298.15)}, \quad K_O = K_{O25} Q_{10,KO}^{0.1(T-298.15)}, \quad R_d = R_{d,25}\,2^{(T_1-298.15)/10}, \tag{22}$$

and A_c, A_j the smaller roots of

$$A_c^2 - \big(V_{c\max} + g(C_i + K_C(1+O_a/K_O)) - R_d\big)A_c + g\big(V_{c\max}(C_i-\Gamma^*)^+ - (C_i + K_C(1+O_a/K_O))R_d\big) = 0, \tag{23}$$

$$A_j^2 - \big(\tfrac{1}{4}J - R_d + g(C_i + 2\Gamma^*)\big)A_j + \tfrac{1}{4}gJ(C_i-\Gamma^*)^+ - g(C_i+2\Gamma^*)R_d = 0, \tag{24}$$

($-R_d$ if the discriminant is negative) and $A_p = 1.5 V_{c\max} - R_d$.

2.4 Stomata and transpiration

With $q_2 = P_s(T_1)$ the saturated vapour pressure at the canopy and the day/night switch $q_7 = \big(1+e^{(\mathrm{PAR}_p-\delta_1)/\gamma_5}\big)^{-1}$ (≈ 1 by day),

$$q_8 = \delta_5 + (\delta_4-\delta_5)q_7, \quad q_9 = \delta_7 + (\delta_6-\delta_7)q_7, \quad q_{10} = \frac{\mathrm{PAR}_p + \delta_2}{\mathrm{PAR}_p + \delta_3}, \tag{25}$$

$$q_3 = \gamma_4\,q_{10}\left[1 + q_8(C_1T_2/536.4 - 200)^2\right]\left[1 + q_9(q_2 - V_1)^2\right] \quad \text{(stomatal resistance, } \mathrm{CO_2} \text{ in ppm)}, \tag{26}$$

$$q_1 = \frac{2\rho_3\alpha_5 L}{\gamma\gamma_2(\gamma_3+q_3)} \quad (q_2 \ge V_1), \qquad q_1 = \frac{2\rho_3\alpha_5 L}{\gamma\gamma_2\gamma_3} \quad (q_2 < V_1,\ \text{dew}), \tag{27}$$

transpiration $p_1 = q_1(q_2 - V_1)$ [$\mathrm{kg\,m^{-2}\,s^{-1}}$] and its latent heat $l_1 = \gamma_2 p_1$.

2.5 Cover node

The cover has no heat capacity. Linearising its long-wave exchange with the crop air and the sky, $k_R = 4\varepsilon_3\sigma\big(\tfrac{1}{2}(T_2 + T_{\mathrm{sky}})\big)^3$,

$$T_c^{(0)} = \frac{k_R(T_2+T_{\mathrm{sky}}) + h_{\mathrm{out}}T_o + Q_{\mathrm{cov}}}{2k_R + h_{\mathrm{out}}}, \qquad T_c = T_c^{(0)} + \frac{\gamma_2\,f_{\mathrm{cov}}\,6.4\times10^{-9}\cdot 1.7\,|T_2 - T_c|^{0.33}\,(V_1 - P_s(T_c))^+}{2k_R + h_{\mathrm{out}}}, \tag{28}$$

the second equation solved by four fixed-point passes (latent heat of the vapour condensing on the cover warms it), with $f_{\mathrm{cov}} = 1 - U_1(1-p_{\mathrm{scr}})$ the share of crop air in contact with the cover (p_{scr} = `vent_scr_perm`).

2.6 Long-wave radiation

With $g_1 = 0.49(1-e^{-\beta_3 L})$, $g_2 = 1-U_1(1-\tau_3)$, $g_3 = \tau_1 g_2(1-0.49\pi\gamma_1\phi_1)e^{-\beta_3 L}$, $F = \varepsilon_3\sigma(T_2^4 - T_c^4)$:

$$r_6 = \alpha_3\varepsilon_1\varepsilon_2\, g_1\, \sigma(T_p^4 - T_1^4) \qquad \text{pipe} \to \text{canopy} \tag{29}$$

$$r_7 = \frac{g_1}{0.49}\varepsilon_2\varepsilon_3\, g_2\, \sigma(T_1^4 - T_c^4) \qquad \text{canopy} \to \text{cover} \tag{30}$$

$$r_{11} = \varepsilon_4\, g_3\, \varepsilon_3\sigma(T_5^4 - T_c^4) \qquad \text{floor} \to \text{cover} \tag{31}$$

$$r_{12} = \varepsilon_{\text{scr}}\, U_1\tau_2\, F \qquad \text{energy screen (at air } T) \to \text{cover} \tag{32}$$

$$r_{13} = A_{\text{cov}}\, \varepsilon_6\, F \qquad \text{air} \to \text{cover} \tag{33}$$

$$r_{14} = \varepsilon_4\varepsilon_2(1-0.49\pi\gamma_1\phi_1)\, \sigma(T_1^4 - T_5^4) \qquad \text{canopy} \to \text{floor} \tag{34}$$

2.7 Air exchange

Roof vents: the leeward and windward sides are combined into one equivalent opening,

$$B = \omega_1\nu_6\frac{T_2 - T_o}{T_2 + T_o}, \quad w = n_3 v^2, \quad n_3 = \nu_2(1-\eta_{11}U_5), \quad \Phi = \tfrac{1}{2}\Big[U_{\text{lee}}^e\sqrt{(B + c_l w)^+} + g_w U_{\text{win}}^e\sqrt{(B + c_w w)^+}\Big], \tag{35}$$

$$U_8 = \Phi/\sqrt{(B+w)^+} \quad \big(\text{if } (B+w)^+ > 10^{-12};\ \text{else } \tfrac{1}{2}(U_{\text{lee}}^e + g_w U_{\text{win}}^e)\big), \tag{36}$$

$(e, g_w, c_l, c_w =$ `vs_exp, vs_gwind, vs_clee, vs_cwind`$)$. Air exchange rates $[\text{m}^3\,\text{m}^{-2}\,\text{s}^{-1}]$:

$$f_7 = \frac{U_8\, f_{\text{scr}}\,\nu_5 n_1}{2\alpha_6}\sqrt{(B+w)^+}, \quad f_{\text{scr}} = 1 - U_1(1-p_{\text{scr}}), \quad n_1 = \nu_1(1-\eta_{10}U_5) \qquad \text{roof vents} \tag{37}$$

$$f_4 = \eta_6 f_7, \qquad f_2 = \eta_6 U_1 f_5, \quad f_5 = \frac{n_1\nu_3 U_6\, v\sqrt{n_3}}{2\alpha_6} \qquad \text{roof, side vents} \tag{38}$$

$$f_3 = U_7\phi_8/\alpha_6 \qquad \text{forced ventilation} \tag{39}$$

$$f_6 = \nu_4\sqrt{\max(v, 0.25)^2 + c_s|T_2 - T_o|} \qquad \text{leakage (wind + stack)} \tag{40}$$

The same total $f_2 + f_3 + f_4 + f_6$ carries heat, vapour and CO_2.

2.8 Heating

Where the logged pipe is warm ($T_{\log} > T_2 + 1$) it is used as is; elsewhere a PI controller requests

$$e = U_{11} - T_2, \quad \Delta = K_p e + K_i \max(e_{\text{int}}, 0), \quad T_p = \max\big(T_{\log},\, T_2 + \text{clamp}(\Delta, 0, T_{\max} - T_2)\big), \tag{41}$$

and $\dot{e}_{\text{int}} = e$ except when the logged pipe is warm, the request saturates with $e > 0$, or $e_{\text{int}} \le 0$ with $e < 0$ (then 0). Pipe-to-air convection: $h_4 = n_p\, 1.99\pi\phi_1\gamma_1|T_p - T_2|^{0.32}(T_p - T_2)$.

2.9 Lamps

The full electrical power $Q_l = P_{\text{el}}U_{12}$ is released inside: $Q_{lc} = f_{\text{can}}(1 - e^{-\beta_1 L})Q_l$ to the canopy, $Q_{la} = Q_l - Q_{lc}$ to the air ($P_{\text{el}} = 117\,\text{W}\,\text{m}^{-2}$, $f_{\text{can}} =$ `lamp_fcan`).

2.10 Sensible and latent heat of the air

$$h_1 = 2\alpha_4 L(T_1 - T_2) \qquad \text{canopy} \to \text{air} \tag{42}$$

$$h_2 = \frac{U_2\phi_7}{\alpha_6}\big(\rho_3\alpha_5 T_o - \gamma_2\rho_3\eta_5(\phi_5 - \phi_6)\big) \qquad \text{fan-pad} \tag{43}$$

$$h_7 = \rho_3\alpha_5(f_2 + f_4 + f_3 + f_6)(T_2 - T_o) \qquad \text{ventilation and leakage} \tag{44}$$

$$h_{11} = 2\nu_7(T_2 - T_5)/(\phi_2 + \nu_8) \qquad \text{air} \to \text{floor conduction} \tag{45}$$

$$Q_{af} = k_{\text{ground}}(T_2 - T_5), \quad Q_{am} = h_m(T_2 - T_m) \qquad \text{air} \to \text{floor, mass convection} \tag{46}$$

$$l_2 = \gamma_2 U_9\phi_9/\alpha_6 \qquad \text{fog evaporation} \tag{47}$$

2.11 Vapour

$$p_2 = \rho_3\frac{U_2\phi_7}{\alpha_6}\big(\eta_5(\phi_5 - \phi_6) + \phi_6\big), \quad p_3 = \frac{U_9\phi_9}{\alpha_6}, \quad p_4 = \frac{\eta_{12}U_4\lambda_4}{\alpha_6}, \quad p_6 = \frac{U_2\phi_7}{\alpha_6}\frac{\psi_1}{\omega_2}\frac{V_1}{T_2}, \tag{48}$$

$$p_5 = \frac{\psi_1}{\omega_2}\Big(\frac{V_1}{T_2} - \frac{V_o}{T_o}\Big)(f_2 + f_3 + f_4 + f_6), \qquad p_7 = U_3\,6.4\times10^{-9}(V_1 - P_s(T_{\text{mc}}))^+, \tag{49}$$

$$p_8 = 6.4\times10^{-9}\cdot 1.7\,U_1|T_2 - T_{\text{scr}}|^{0.33}(V_1 - P_s(T_{\text{scr}}))^+ \qquad \text{screen condensation} \tag{50}$$

$$p_9 = 6.4\times10^{-9}\cdot 1200\,|T_1 - T_2|^{0.33}(V_1 - P_s(T_2))^+ \qquad \text{supersaturated air} \tag{51}$$

$$p_{10} = A_{\text{cov}}f_{\text{cov}}\,6.4\times10^{-9}\cdot 1.7\,|T_2 - T_c|^{0.33}(V_1 - P_s(T_c))^+ \qquad \text{cover condensation} \tag{52}$$

2.12 CO_2

$$o_1 = \frac{\eta_{13}U_4\lambda_4}{\alpha_6},\ o_2 = \frac{U_{10}\psi_2}{\alpha_6},\ o_3 = \frac{U_2\phi_7}{\alpha_6}(C_1 - C_o),\ o_5 = (C_1 - C_o)(f_2 + f_3 + f_4 + f_6),\ o_R = R_{\text{crop}}\,2^{(T_1 - 293.15)/10}. \tag{53}$$

2.13 Floor, deep soil and internal mass

$Q_{fd} = k_{\text{deep}}(T_5 - T_d)$; the floor balance (7) collects air convection and conduction, canopy long-wave, its own long-wave loss to the cover and its share of the solar column. The internal mass (structure, slabs, water, standing crop) exchanges Q_{am} with the air and absorbs Q_{ms}.

2.14 Psychrometrics

$P_s(T) = 611.21\exp\big[(18.678 - \frac{T_c}{234.5})\frac{T_c}{257.14 + T_c}\big]$ above 0 °C (ice form below), $T_c = T - 273.15$; $\text{RH} = 100V_1/P_s(T_2)$.

3 Crop model: Aaron Vélez's generic cucumber model with canopy and LAI

The crop model is the generic dry-matter partitioning model of A. Vélez (`Generic_Gh_Model`: `models.py`, `main.tex`), which casts Marcelis (1994, "M94") for cucumber into a generic form, plus his canopy extension (leaf mass, LAI, retained leaves, pruning). Code: `src/crop.jl` (`CucumberGeneric`, `crop_step!`) and `src/crop_measured.jl`. States are per m^2 floor; one stem is one modelled plant ($\rho = 3.6$ stems m^{-2}, Hemming et al. 2019). Day k (0 = transplant, 14 August), daily mean air temperature T [°C].

3.1 States

symbol	code	meaning
W_L, W_S, W_R	`W_leaf, W_stem, W_root`	leaf, stem, root dry mass [g m^{-2}]
L	`LAI`	leaf area index
N_L	`leaves`	retained leaves per stem
C_N	`node`	node counter per stem (fractional progress kept)
(z_i, W_i, n_i)	`Fruit(dev, W, n, day_set)`	fruit cohort: stage, dry mass [g m^{-2}], number [m^{-2}]
R	`reserve`	carbohydrate reserve (DM equivalent) [g m^{-2}]
Y_{DW}, Y_{FW}	`yield_DW, yield_FW`	cumulative harvest [g DM m^{-2}], [kg FW m^{-2}]

3.2 Generic sub-models (M94 in generic form)

$$a(T) = \max(\beta_0 + \beta_1 T, 0)\, f_{\mathrm{PAR}}, \quad \beta_0 = -\beta_1 \tfrac{0.75}{0.09}, \quad f_{\mathrm{PAR}} = \frac{1 - e^{-0.5 - 0.5\,\mathrm{PAR}_k}}{1 - e^{-0.5 - 0.5\cdot 6.8}} \quad \text{nodes stem}^{-1}\text{d}^{-1} \tag{54}$$

$$\dot{z} = \kappa \max(T - 10, 0), \qquad \kappa = 1/E_{\mathrm{fruit}} \quad \text{fruit development} \tag{55}$$

$$P_i = n_i\, \max(A_0 + A_T T, 0)\, \frac{u}{(1+u)^p}, \quad u = \frac{e^{-s(\min(z_i,1) - z_{\mathrm{peak}})}}{p-1} \quad \text{fruit sink [g d}^{-1}\text{m}^{-2}\text{]} \tag{56}$$

$$P_V = \rho \max\big(V_{20} + V_T(T - 20), 0\big) \quad \text{vegetative demand} \tag{57}$$

with $s = 0.0170 \cdot 275$, $p = 1 + 1/0.0111$, $z_{\mathrm{peak}} = 131/275$, $V_T = 0.25$, PAR_k the day's canopy-top PAR [MJ m^{-2} d^{-1}]. The sink amplitude is parametrised by the potential fruit dry mass,

$$A_T = \frac{W_{\mathrm{pot}}\, s\, (p-1)}{E_{\mathrm{fruit}}}, \quad A_0 = -10 A_T \quad \Rightarrow \quad \int_0^1 P_i \frac{dz}{\dot{z}} \simeq n_i W_{\mathrm{pot}}, \tag{58}$$

so W_{pot} sets the final fruit size and E_{fruit} the duration. A fruit past $z = 1$ keeps the sink of $z = 1$.

3.3 Leaf area index

LAI is a state driven by leaf mass and leaf events; there is no maximum LAI, no recomputation of the standing area and no senescence; all retained leaves share the canopy's mean mass and area.

- *Leaf events.* Every completed node adds a leaf: $N_L \leftarrow N_L + \lfloor C_{N,k+1} \rfloor - \lfloor C_{N,k} \rfloor$.
- *Area growth.* $\Delta W_L = f_L \Delta W_V$, $\Delta L = S(d_k) \Delta W_L$ with $f_L = 0.59$ and the seasonal SLA

$$S(d) = 10^{-4}\Big[266 + 88 \sin \frac{2\pi(d + 68)}{365}\Big]\ \mathrm{m}^2\,\mathrm{g}^{-1}, \qquad d_k = 226 + k, \tag{59}$$

  (Heuvelink 1999, a tomato regression used provisionally), applied only to new leaf mass.
- *Pruning.* If $N_L > N_{\mathrm{target}}$: $r = N_{\mathrm{target}}/N_L$, $W_L \leftarrow r W_L$, $L \leftarrow rL$, $N_L \leftarrow N_{\mathrm{target}}$ (removed mass and area to a ledger; non-leaf mass untouched).
- *Transplant.* $L_0 = 0.25$, $C_N = N_0$, $N_L = \lfloor N_0 \rfloor$, $W_L = L_0/S(226)$, $W_S + W_R = W_L(1 - f_L)/f_L$ split $0.32 : 0.08$.

3.4 Daily step

1. **Supply.** $R_m = Q_{10}^{(T-25)/10}(k_{m,L} W_L + k_{m,S} W_S + k_{m,R} W_R + k_{m,F} \sum_i W_i)$, $B = \max(P_g - R_m, 0)/c$, $c = 1.4$.
2. **Allocation** with reserve: $\Delta W_j = \frac{P_j}{P} \min(B + R, P)$, $P = P_V + \sum_i P_i$, $R \leftarrow B + R - \sum_j \Delta W_j$.

3. **Vegetative growth and LAI**: as above; non-leaf mass split stem:root = 0.32 : 0.08; $W_i \mathrel{+}= \Delta W_i$.
4. **Development and appearance** (before topping, day 104): $z_i \mathrel{+}= \dot{z}$, $C_N \mathrel{+}= a(T)$; nodes ≥ 5 carry a flower; from day 15 the flowers set $n_{\text{new}} = n_{\text{fl}}\,\rho\,\phi_k\,\text{clamp}\big(\frac{B/P - r_{\text{low}}}{r_{\text{high}} - r_{\text{low}}}, 0, 1\big)$ fruits (new cohort at $z = 0$, growing from the next day), $\phi_k = 1 - \text{Pruning}_k/100$ the grower's kept share.
5. **Harvest** when $W_i/n_i \geq 323\,\text{DMC}_{\text{fruit}}$ (observed mean A+B fruit, 323 g FW) or $z_i \geq 3$.
6. **Pruning** to N_{target}.

3.5 Light and photosynthesis of the crop path

Gross assimilation P_g [g CH_2O m^{-2} d^{-1}] is the FvCB model of Section 2.3 summed over the 5-min samples of the day, at measured air T and CO_2, with the organisers' canopy-top PAR

$$\text{PAR} = \big[\text{PAR}_{\text{out}} \cdot 0.65\,(1 - 0.98U_{\text{bl}})(1 - 0.25U_1) + 187\,U_{12}\big]/4.6 \quad [\text{W m}^{-2}], \tag{60}$$

multiplied by the both-faces factor $g(L) = (D_4^P + \rho_2 D_5^P)/t_r$ of the PAR column with screens retracted, effective leaf area $(1 - e^{-0.75L})/0.75$ and constant R_d.

3.6 Climate inputs from the crop

`crop_series`: L; $R_{\text{crop}} = \big[2^{-0.5}(k_{m,S}W_S + k_{m,R}W_R + k_{m,F}\sum W_i) + (c-1)\Delta DM\big]\frac{44}{30}\frac{1000}{86400}$ (non-leaf maintenance at 20 °C + growth respiration; leaf dark respiration is in the FvCB model); $\text{FW} = (W_S/0.06 + W_R/0.07 + \sum W_i/\text{DMC}_{\text{fruit}})/1000$.

4 Parameters

Table 2: Climate and photosynthesis parameters (`configfiles/climate_n79.json`). Inferred parameters: [min, centre, max] of the uniform prior; the centre is the default value.

name	value (prior)	units	meaning; source
geometry			
`alpha3`	0.3	m^2 m^{-2}	pipe surface seen by the canopy (long-wave); V2.0 (audit: geometric 0.16)
`alpha6`	10000	m^2	floor area used to express absolute capacities per m^2 floor; V2.0
`cover_area`	1.6	m^2 m^{-2}	exterior cover area per floor area (roof + one side wall); Note 67
`gamma1`	1	m m^{-2}	heating-pipe length per floor area (rail pipe, one circuit); Note 78
`n_pipes`	1	-	number of heating circuits (crop pipe folded into the rail pipe by the inputs); Note 54/79
`nu3`	2352.9	m^2	side-wall vent surface (side vents, input U6); V2.0
`nu5`	3000	m^2	roof-vent area (0.30 of the floor, insect nets); Notes 78, 92
`nu6`	1	m	vertical dimension of a roof-vent opening; V2.0
`nu8`	0.64	m	floor-to-ground distance in the air-floor conduction length; V2.0
`phi1`	0.051	m	outer diameter of the heating pipe; V2.0 / Note 78
`phi2`	5.4	m	mean air height (volume per floor area); Note 67 (compartment drawing)

name	value (prior)	units	meaning; source
constants			
alpha5	1005	J kg^{-1} K^{-1}	specific heat of air; Note 78
gamma	65.8	Pa K^{-1}	psychrometric constant; V2.0
gamma2	2.45×10^6	J kg^{-1}	latent heat of evaporation; V2.0
omega1	9.81	m s^{-2}	gravity; V2.0
omega2	8314	J $kmol^{-1}$ K^{-1}	molar gas constant; V2.0
psi1	18	kg $kmol^{-1}$	molar mass of water; V2.0
rho3	1.2	kg m^{-3}	air density; V2.0
sigma	5.67×10^{-8}	W m^{-2} K^{-4}	Stefan-Boltzmann constant; V2.0
air			
air_cap_mult	**inferred** [1, 3, 12]	-	effective air heat-capacity multiplier (fast masses lumped with the air); n79 prior (Note 78; f_mass Note 104)
canopy			
alpha1	1500	J K^{-1} m^{-2} leaf	canopy heat capacity per leaf area; V2.0
alpha4	5	W m^{-2} K^{-1}	leaf-air convective coefficient (per leaf side); Vanthoor 2011
beta1	0.7	-	canopy PAR extinction coefficient; V2.0
beta3	0.27	-	canopy NIR extinction coefficient (also used for the long-wave terms); V2.0 (audit: K_FIR 0.94)
delta1	5	W m^{-2}	PAR defining day/night for the stomatal resistance; V2.0
delta2	4.3	W m^{-2}	stomatal light response parameter; V2.0
delta3	0.54	W m^{-2}	stomatal light response parameter; V2.0
delta4	6.1×10^{-7}	ppm^{-2}	stomatal CO2 response, day; V2.0
delta5	1.1×10^{-11}	ppm^{-2}	stomatal CO2 response, night; V2.0
delta6	4.3×10^{-6}	Pa^{-2}	stomatal vapour-pressure response, day; V2.0
delta7	5.2×10^{-6}	Pa^{-2}	stomatal vapour-pressure response, night; V2.0
epsil2	1	-	canopy long-wave emissivity; V2.0
gamma3	275	s m^{-1}	canopy boundary-layer resistance to vapour; V2.0
gamma4	**inferred** [40, 82, 200]	s m^{-1}	minimum stomatal resistance; n79 prior (Note 78; f_mass Note 104)
gamma5	-1	W m^{-2}	inverse slope of the day/night switch of the stomatal resistance; V2.0
k_lai_photo	0.7	-	extinction in the effective leaf area (1-exp(-kL))/k of the climate photosynthesis; V2.2 K_EXT
rho1	0.07	-	canopy PAR reflection at high LAI; V2.0
rho_can_nir	0.35	-	canopy NIR reflection at high LAI; Vanthoor 2011 / Note 103
photosynthesis			
E_Soc	-24460	J mol^{-1}	activation energy of Sco; V2.0
J_max	2×10^{-4}	mol m^{-2} s^{-1}	maximum electron transport rate (climate model; the crop calibration infers its own); V2.0
K_C25	3×10^{-4}	mol m^{-3}	Michaelis constant for CO2 at 25 C; V2.0
K_O25	0.3	mol m^{-3}	Michaelis constant for O2 at 25 C; V2.0
O_a	0.21	mol m^{-3}	O2 concentration; V2.0
Q10_KC	2.1	-	Q10 of K_C; V2.0
Q10_KO	1.2	-	Q10 of K_O; V2.0
Q10_Vcmax	2.4	-	Q10 of Vcmax; V2.0
Rd_day	1×10^{-6}	mol m^{-2} s^{-1}	leaf dark respiration at 25 C (x Q10 2 on canopy T); V2.0 / Note 86

name	value (prior)	units	meaning; source
Rgas	8.31447	J mol^{-1} K^{-1}	gas constant; V2.0
Sco25	2800	-	Rubisco CO2/O2 specificity at 25 C; V2.0
V_cmax25	1.5×10^{-4}	mol m^{-2} s^{-1}	maximum carboxylation rate at 25 C; V2.0
alpha	0.35	-	quantum efficiency of electron transport; V2.0
g_m	0.25	mol m^{-2} s^{-1}	mesophyll conductance in the constant effective conductance; Bush 2023 (V2.0)
g_s	0.3	mol m^{-2} s^{-1}	stomatal conductance in the constant effective conductance; Bush 2023 (V2.0)
theta	0.7	-	curvature of the light response; V2.0
floor			
C_soil	**inferred** [1.5×10^5, 6×10^5, 2×10^6]	J m^{-2} K^{-1}	heat capacity of the active floor layer; n79 prior (Note 78; f_mass Note 104)
alpha7	0.5	-	floor NIR absorption (reflection 1 - alpha7); V2.0
epsil4	1	-	floor long-wave emissivity; V2.0
k_deep	**inferred** [0.5, 1.5, 3]	W m^{-2} K^{-1}	floor to deep-soil conductance; n79 prior (Note 78; f_mass Note 104)
k_ground	**inferred** [1, 8, 40]	W m^{-2} K^{-1}	air-floor convective conductance; n79 prior (Note 78; f_mass Note 104)
nu7	0.85	W m^{-1} K^{-1}	soil conductivity in the air-floor conduction h11; V2.0
rho2	0.65	-	floor PAR reflection; V2.0
sd_tau	1.2096×10^6	s	time constant of the deep-soil temperature; Note 47
mass			
C_mass	**inferred** [20000, 50000, 1.2×10^5]	J m^{-2} K^{-1}	heat capacity of the internal mass (structure, slabs, water); n79 prior (Note 78; f_mass Note 104)
crop_cp	3800	J kg^{-1} K^{-1}	specific heat of fresh plant tissue; Note 91
f_mass	**inferred** [0, 0.3, 1]	-	share of the floor-level solar absorbed by the internal mass; n79 prior (Note 78; f_mass Note 104)
h_mass	**inferred** [1, 5, 40]	W m^{-2} K^{-1}	air-internal mass heat-transfer coefficient; n79 prior (Note 78; f_mass Note 104)
cover			
cov_abs	0.1	-	share of the global radiation absorbed by the glass (booked to the cover); Note 46
epsil3	1	-	sky/cover long-wave emissivity; V2.0
epsil6	0.44	-	cover long-wave emissivity (air-cover term r13); V2.0
eta1	0.1	-	share of the global radiation absorbed by the structure (booked to the air); V2.0
eta2	0.5	-	PAR share of the global radiation; V2.0
eta3	0.5	-	NIR share of the global radiation; V2.0
h_cover_out	**inferred** [10, 25, 50]	W m^{-2} K^{-1}	cover-outside convective coefficient; n79 prior (Note 78; f_mass Note 104)
tau1	0.722222	-	cover PAR transmission; (1-eta1) tau1 = 0.65 roof transmission; Note 78 (organisers 0.60-0.70)
tau2	1	-	cover long-wave transmission in the screen term r12; V2.0
screens			
eps_screen	0.4	-	energy-screen long-wave emissivity (upper face, r12); V2.2
scr_abs_air	0.5	-	share of screen-absorbed solar released to the crop air (rest to the cover); Note 101 (PROVISIONAL)

name	value (prior)	units	meaning; source
scr_bl_rho_nir	0.5	-	blackout screen NIR reflection (PROVISIONAL, = PAR); Note 101
scr_bl_rho_par	0.5	-	blackout screen PAR reflection (PROVISIONAL, assumed); Note 101
scr_bl_tau_nir	0.02	-	blackout screen NIR transmission (PROVISIONAL, = PAR); Note 101
scr_bl_tau_par	0.02	-	blackout screen PAR transmission at full closure (PROVISIONAL); organisers 98 % interception
scr_es_rho_nir	0.15	-	energy screen NIR reflection (PROVISIONAL, = PAR); Note 101
scr_es_rho_par	0.15	-	energy screen PAR reflection (PROVISIONAL, assumed); Note 101
scr_es_tau_nir	0.75	-	energy screen NIR transmission (PROVISIONAL, = PAR); Note 101
scr_es_tau_par	0.75	-	energy screen PAR transmission at full closure (PROVISIONAL); organisers 25 % interception
tau3	0.11	-	energy-screen long-wave transmission; V2.0
heating			
epsil1	0.88	-	pipe long-wave emissivity; V2.0
hc_Ki	0.00555556	s^{-1}	PI heating controller integral gain (tau_i 30 min); Note 34
hc_Kp	10	K K^{-1}	PI heating controller proportional gain; Note 34
hc_Tmax	338.15	K	maximum pipe temperature (65 C); Note 33
ventilation			
eta10	0	-	shading effect on the discharge coefficient; V2.0
eta11	0	-	shading effect on the wind-pressure coefficient; V2.0
eta6	0.551	-	ventilation reduction factor (Vanthoor 8.70, zeta 0.33); V2.0
leak_cs	**inferred** [0, 0.3, 1]	m^2 s^{-2} K^{-1}	stack (temperature-difference) term of the leakage; n79 prior (Note 78; f_mass Note 104)
nu1	**inferred** [0.3, 0.65, 1]	-	roof-vent discharge coefficient; n79 prior (Note 78; f_mass Note 104)
nu2	**inferred** [0.005, 0.05, 0.3]	-	roof-vent wind-pressure coefficient; n79 prior (Note 78; f_mass Note 104)
nu4	**inferred** [1×10^{-5}, 4.5×10^{-5}, 1×10^{-4}]	m s^{-1} per m s^{-1}	leakage coefficient; n79 prior (Note 78; f_mass Note 104)
vent_scr_perm	**inferred** [0, 0.4, 1]	-	share of roof ventilation and cover contact reaching the crop air through a closed energy screen; n79 prior (Note 78; f_mass Note 104)
vs_clee	1	-	leeward wind-pressure multiplier; Note 58
vs_cwind	**inferred** [0.5, 1, 10]	-	windward wind-pressure multiplier; n79 prior (Note 78; f_mass Note 104)
vs_exp	**inferred** [0.5, 0.9, 1.2]	-	curvature of opening -¿ flow; n79 prior (Note 78; f_mass Note 104)
vs_gwind	**inferred** [0.5, 1.5, 4]	-	windward discharge multiplier relative to leeward; n79 prior (Note 78; f_mass Note 104)
lamps			
lamp_Pel	117	W m^{-2}	lamp electrical power at 100 % (187 umol / 1.6 umol J^{-1}); Note 90
lamp_fcan	**inferred** [0.05, 0.3, 0.9]	-	share of lamp power absorbed by a closed canopy; n79 prior (Note 78; f_mass Note 104)

name	value (prior)	units	meaning; source
`lamp_par_W`	40.65	W m^{-2}	lamp PAR at the canopy at 100 % (187 umol / 4.6); Note 71
actuators			
`eta12`	4.43×10^{-8}	kg J^{-1}	vapour released per J of direct-heater heat (U4); V2.0
`eta13`	0.057	mg J^{-1}	CO2 released per J of direct-heater heat (U4); V2.0
`eta5`	0.5	-	fan-pad efficiency (U2); V2.0
`lamb4`	5×10^5	W	direct air heater capacity (U4); V2.0
`phi5`	0.014	kg kg^{-1}	vapour content of fan-pad air (U2); V2.0
`phi6`	0.0079	kg kg^{-1}	vapour content of outside air for the fan-pad (U2); V2.0
`phi7`	16.7	m^3 s^{-1}	fan-pad air flow capacity (U2); V2.0
`phi8`	666.6	m^3 s^{-1}	forced-ventilation capacity (U7); V2.0
`phi9`	0.916	kg s^{-1}	fog capacity (U9); V2.0
`psi2`	35556	mg s^{-1}	CO2 valve capacity (12.8 g m^{-2} h^{-1} x alpha6); Note 78
noise			
`sigma_C`	**inferred** [10, 60, 400]	mg m^{-3}	likelihood noise; n79 prior
`sigma_RH`	**inferred** [0.5, 6, 25]	%	likelihood noise; n79 prior
`sigma_T`	**inferred** [0.2, 1.5, 6]	K	likelihood noise; n79 prior

Table 3: Crop parameters (`configfiles/crop_n105.json`).

name	value (prior)	units	meaning; source
M94			
`E_fruit`	**inferred** [150, 275, 450]	C d	fruit duration (degree-days above 10 C); n94/n105 prior (Notes 74, 94, 98)
`V20`	**inferred** [1, 3.2, 8]	g DM $stem^{-1}$ d^{-1}	vegetative demand at 20 C; n94/n105 prior (Notes 74, 94, 98)
`VT`	0.25	g DM $stem^{-1}$ d^{-1} C^{-1}	temperature slope of vegetative demand; Marcelis 1994
`W_pot`	**inferred** [20, 60.7, 120]	g DM $fruit^{-1}$	potential fruit dry mass; n94/n105 prior (Notes 74, 94, 98)
`app_base`	8.33333	C	base temperature of node appearance (beta0 = -beta1 x 0.75/0.09); Marcelis 1994
`app_beta1`	**inferred** [0.05, 0.0882, 0.13]	nodes $stem^{-1}$ d^{-1} C^{-1}	node (flower) appearance slope; n94/n105 prior (Notes 74, 94, 98)
`dev_base`	10	C	base temperature of fruit development and of the sink amplitude; Marcelis 1994
`par_ref`	6.8	MJ m^{-2} d^{-1}	PAR at which M94 fitted the appearance rate; Marcelis 1994
`sink_p`	91.0901	-	fruit sink exponent p = 1 + 1/0.0111; Marcelis 1994 via Aaron
`sink_peak`	0.476364	-	stage of peak sink = 131/275; Marcelis 1994 via Aaron
`sink_s`	4.675	-	fruit sink shape s = 0.0170 x 275; Marcelis 1994 via Aaron
canopy			
`LAI0`	0.25	m^2 m^{-2}	LAI at transplant (assumed); Note 94
`frac_root`	0.08	-	root weight in the stem:root split; V2.2
`frac_stem`	0.32	-	stem weight in the stem:root split; V2.2

name	value (prior)	units	meaning; source
gen_N0	**inferred** [2, 4, 8]	nodes	nodes at transplant; n94/n105 prior (Notes 74, 94, 98)
gen_doy0	226	day of year	transplant day (14 Aug); Note 94
gen_fL	0.59	-	leaf share of vegetative growth; Aaron (Note 94)
gen_leaf_target	**inferred** [20, 25, 30]	leaves stem^{-1}	retained-leaf target (pruning); n94/n105 prior (Notes 74, 94, 98)
sla_amp	88	cm2 g^{-1}	seasonal SLA amplitude; Heuvelink 1999 eq. 3
sla_mean	266	cm2 g^{-1}	seasonal SLA mean; Heuvelink 1999 eq. 3
sla_phase	68	d	seasonal SLA phase; Heuvelink 1999 eq. 3
stem_density	3.6	stems m^{-2}	stem density (one stem = one modelled plant); Hemming et al. 2019 (Note 92)
architecture			
gen_flower_node	5	node	first node with a flower; Aaron (Note 94)
set_start_day	15	d	first day with fruit set; Note 66
top_day	104	d	topping: no node appearance after this day; V2.2
respiration			
Q10_resp	2	-	Q10 of maintenance; V2.2
asrq	1.4	g CH2O g^{-1} DM	construction cost; Aaron (1.4)
k_m_fruit	**inferred** [0.003, 0.01, 0.02]	g g^{-1} d^{-1}	fruit maintenance at 25 C; n94/n105 prior (Notes 74, 94, 98)
k_m_leaf	**inferred** [0.01, 0.03, 0.06]	g g^{-1} d^{-1}	leaf maintenance at 25 C; n94/n105 prior (Notes 74, 94, 98)
k_m_root	0.01	g g^{-1} d^{-1}	root maintenance at 25 C; V2.2
k_m_stem	**inferred** [0.005, 0.015, 0.03]	g g^{-1} d^{-1}	stem maintenance at 25 C; n94/n105 prior (Notes 74, 94, 98)
fruit set			
set_r_high	**inferred** [0.2, 1, 2.5]	-	source/sink ratio with full fruit set; n94/n105 prior (Notes 74, 94, 98)
set_r_low	**inferred** [0, 0.3, 1]	-	source/sink ratio with no fruit set; n94/n105 prior (Notes 74, 94, 98)
quality			
DMC_fruit	**inferred** [0.025, 0.035, 0.045]	-	fruit dry-matter content; n94/n105 prior (Notes 74, 94, 98)
harvest			
harvest_FW	323	g FW	picking size (observed mean A+B fruit); Note 92
photosynthesis			
J_max	**inferred** [1×10^{-4}, 2×10^{-4}, 3.5×10^{-4}]	mol m^{-2} s^{-1}	maximum electron transport rate (crop path); n94/n105 prior (Notes 74, 94, 98)
k_ext	0.75	-	extinction in the effective leaf area of the crop path; V2.2 (climate uses 0.7)
light			
lamp_umol	187	umol m^{-2} s^{-1}	lamp PAR at 100 %; AGC ReadMe
transmission	0.65	-	greenhouse PAR transmission in the measured-light formula; Note 78
series			
dmc_root	0.07	-	root dry-matter content (standing fresh mass); Note 91 (assumed)
dmc_stem	0.06	-	stem dry-matter content (standing fresh mass); Note 91 (assumed)
noise			

name	value (prior)	units	meaning; source
`sigma_fd`	**inferred** [0.5, 3, 15]	-	likelihood noise; n94/n105 prior (Notes 74, 94, 98)
`sigma_leaf`	**inferred** [0.5, 3, 20]	-	likelihood noise; n94/n105 prior (Notes 74, 94, 98)
`sigma_n`	**inferred** [0.3, 3, 20]	-	likelihood noise; n94/n105 prior (Notes 74, 94, 98)
`sigma_y`	**inferred** [0.02, 0.3, 3]	-	likelihood noise; n94/n105 prior (Notes 74, 94, 98)

5 Parameter inference

Both calibrations use the t-walk sampler, uniform priors (Tables 2, 3), a Gaussian likelihood with inferred noise per observable,

$$U(\theta, \sigma) = \sum_k \Big[n_k \log \sigma_k + \frac{\mathrm{SSE}_k(\theta)}{2\sigma_k^2} \Big], \tag{61}$$

two or more seeds, split-$\hat{R}$ and an explicit burn-in.

- **Crop** (`test/calibrate_crop.jl`): 13 parameters + 4 noise terms; observations weekly yield, weekly harvested fruit number, leaves per stem, fruit development time; driver measured climate.
- **Climate** (`test/calibrate_climate.jl`): 18 parameters + 3 noise terms; 7-day blocks with a spin-up day and the measured initial state; TRAIN days 7–13, 28–34, 63–69, 91–97, 110–114; VAL 42–48, 77–83; TEST the rest; observations hourly means of air T and RH and dark-hour CO_2; vents, pipe, screens, lamps and CO_2 valve are logged inputs.

What to infer and what to fix. Measurable or documented quantities are fixed (geometry, materials, actuator capacities, crop counts and dates); missing values of this kind are questions for the organisers or data sheets. Only coefficients of approximate sub-models are inferred. No bias or measurement-offset terms. As the physics becomes more descriptive, effective parameters should be retired: $h_{\mathrm{out}} \to 2.8 + 1.2v$ with a conserving cover node; `air_cap_mult` → explicit fast masses; `lamp_fcan` → lamp spectrum through the column; `vent_scr_perm` → screen permeability in a top compartment.

6 Known structural gaps

These are properties of the equations above (read-only audit of 21 October 2026 and the project notes); they are the starting list for the physics work.

1. **Cover node not conserving.** The node of Section 2.5 balances $k_R(T_2 - T_c)$ only; it does not receive r_7, r_{11}, r_{12}, r_{13} nor pipe radiation ($40\,\mathrm{W\,m^{-2}}$ residual on a test night). r_{11} contains the PAR transmission τ_1. The node is not scaled by the cover area while r_{13} is.
2. **No top compartment.** The screen sits at $\frac{1}{2}(T_o + T_2)$ for condensation and at T_2 in r_{12}; r_{13} has no screen factor; cover condensation draws on the crop air (through f_{cov}). Geometry: screen $\approx 4.5\,\mathrm{m}$, ridge $\approx 5.8\,\mathrm{m}$, $\approx 0.9\,\mathrm{m^3\,m^{-2}}$ above the screen.
3. **Long-wave network against LAI.** r_{14} has no canopy fraction; β_3 (NIR) is used for long-wave instead of $K_{\mathrm{FIR}} = 0.94$; pipes radiate only to the canopy ($\alpha_3 = 0.3$ vs geometric 0.16), no pipe–floor, pipe–cover terms.
4. **Lamps.** At $f_{\mathrm{can}} = 0.3$, $L = 2.3$ the canopy heat ($28.1\,\mathrm{W\,m^{-2}}$) is below the absorbed lamp PAR (34.3); no lamp light reaches the floor.

5. **Inherited V2.0 forms**: side-vent flow scaled by the energy screen ($f_2 = \eta_6 U_1 f_5$); air–floor conduction length uses the air height ($\phi_2 + \nu_8$); mechanical cooling removes vapour but no heat; h_2 fan-pad form. Zero input in AiCU except h_{11}.
6. **Fit at the prior centre** (V2.2 smoke 4): air ≈ 1.2 K too cold (day -1.6, night -1.0), RH ≈ -10 %.
7. **Photosynthesis**: big leaf; $J_{\max}$ of the climate model (JSON) differs from the crop posterior; $k = 0.7$ (climate) vs 0.75 (crop path); constant R_d in the crop path.
8. **Crop**: no fruit abortion (fruit number 107 vs 116 observed); SLA from tomato; bimodal fruit size/duration posterior.
9. **Provisional optics**: screen reflectances and long-wave properties, NIR = PAR for the screens, $\rho_\infty^{\mathrm{NIR}}$, `scr_abs_air`.
10. **Coupling**: crop → climate only, offline.
11. **Numerical noise of the likelihood** (Section 7): at reltol 10^{-4} the energy changes by up to ∼60 units for a 10^{-14} relative change of one parameter. Smoothing the switches (heating, clamps, condensation onsets), stopping the solver at input steps, or a tighter tolerance are needed before long MCMC runs.

7 Verification

The code was obtained from GreenhouseSim V2.2 (git tag `v2.2-flags-final`) by resolving every flag to its n79 value (`PRIORS=n79`, `SOLARC=1`, `PIPE2=1`, `CO2NIGHT=1`) and the crop to the n105 set-up, and deleting the branches not taken. Four checks support the claim that the code is n79 and that this document is the code:

1. `tools/audit.py`: no flag name or `hasproperty(p, ...)` switch is left in `src/`; every parameter the code reads is in the JSON files and every JSON key is read.
2. `test/equivalence_synthetic.jl` (no data): V2.2 with the n79 flags and GreenhouseTwin on the same synthetic inputs: `rhs!` at 5000 random states and times (relative difference $< 10^{-10}$), 3-day trajectories, solar fluxes, and the crop over 116 synthetic days.
3. `test/equivalence_n79.jl` (AGC data): both codes build their inputs independently from the AGC files (compared array by array), parameters, `rhs!` at random states, TRAIN blocks, likelihood energies at the prior centre and at a jittered point, the full season, the solar budget, the crop at the n105 prior centre, timing and allocations.
4. Term dump: `test/dump_terms.jl` writes every term of `model_terms` at test states; `tools/check_tex.py` evaluates the equations of Section 2 written in Python from this document and compares.

Results (21 October 2026, Julia 1.10). Audit: pass. Synthetic: 35/35. Document against code (`check_tex.py`, 400 states, 60 terms): largest relative difference 2×10^{-14}; this check found and corrected a sign error in the FvCB quadratics of an earlier draft of Section 2.3. AGC data: all 20 input arrays and 112 shared parameters identical; `rhs!` derivatives within 3×10^{-16} relative; solar fluxes and crop series identical (crop to 10^{-14}); `rhs!` 1.7 μs against 2.3 μs for V2.2, no allocations.

Numerical noise of the trajectories and of the likelihood. Because the right-hand side is discontinuous in time (5-min step inputs) and in state (heating switch $T_{\log} > T_2 + 1$, PI clamp, max/min, condensation onsets), the adaptive solver turns rounding-level differences into finite differences of the solution. V2.2 compared with itself after scaling one parameter by $1 + 10^{-14}$:

reltol	TRAIN-block T_2 difference	energy sd (10 perturbations)	energy range
10^{-4} (calibration default)	0.39 K	13–24	37–62
10^{-6}	–	0.36–0.64	1.0–2.1
10^{-8}	a block solve fails (both codes)		

At 10^{-6} the mean energies of V2.2 and this code agree to 0.11 (of 6945) at the prior centre; at a jittered point a difference of 0.55 at 10^{-6} shrinks to 0.035 ± 0.036 at 10^{-7}, i.e. it is integration error, not a model difference. Trajectory and energy checks are therefore made against this noise floor, not against fixed tolerances. For calibration the noise at 10^{-4} is of the order of the energy differences the sampler uses; see Section 6.

The few V2.2 terms that were booked and then cancelled (lamp NIR on the canopy, V2.0 lamp heat on the air, floor long-wave first taken from the air and then returned) are booked once here, so the results agree to rounding rather than bit for bit.

Table 4: Where each part of this document is in the code.

document	code
state equations (2)–(9)	`model_terms`, end of each block; `rhs!` (`src/climate.jl`)
solar column	`solar_column`, `column4`, `canopy_layer`, `roof_props` (`src/solar.jl`)
plant light, FvCB	`PARp`, `assimilation`, `lai_eff` (`src/photosynthesis.jl`)
stomata	`q2`–`q10`, `q1`
cover node	`cover_temperature`
long-wave	`r6`, `r7`, `r11`–`r14`
air exchange	`U8`, `f2`–`f7`
heating	`e`, `dmd`, `Tp`, `deint`, `h4`
lamps	`Ql`, `Qlc`, `Qla`
air heat, vapour, CO_2	`h1`–`h11`, `l1`, `l2`, `p1`–`p10`, `o1`–`oR`
floor, deep soil, mass	`Qaf`, `Qfd`, `Qfs`, `Qam`, `Qms`, `Cm`
inputs	`src/inputs.jl`, `src/agc_data.jl`
crop sub-models	`node_appearance`, `development_rate`, `fruit_sink`, `vegetative_demand`, `cucumber_generic` (`src/crop.jl`)
LAI, daily step	`seasonal_sla`, `crop_step!`, `init_crop_state`
crop light, series	`daily_assimilation_measured`, `load_measured_climate_par`, `crop_series` (`src/crop_measured.jl`)
likelihoods	`climate_energy` (`src/climate_calibration.jl`), `crop_energy`